Nishal Stanislaus Silva, *PhD*

Music Technology ML Research Engineer | Audio, DSP & Real-Time Systems

Canadian Permanent Resident, Eligible to work in Canada without sponsorship

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Professional Summary

Music technology ML research engineer with a PhD focused on real-time audio pattern detection and a research track record built almost entirely on music-content analysis. Designed and published MIRaaS, a UDP edge-to-server offload architecture for Music Information Retrieval as a service (IEEE IS2 2026) directly analogous to the content-analysis infrastructure a music/media platform runs at scale. Created and released four open music-content datasets (DoMP, DoPP, DoDP, DoDP2; 9,800+ recordings, 70+ musicians) and shipped symbolic-plus-audio music modelling (onset/beat, chroma, key/chord, MIDI). Complements this with 15+ years as a performing and session guitarist (Trinity College London Rock & Pop Guitar, Grade 8), giving direct musical-domain intuition alongside the engineering.

Portfolio & Demos

Nebula – C++17 audio-feature-extraction / MIR library with per-feature ARM and Raspberry Pi latency benchmarks: github.com/ns2max/nebula. **"Hot Licks" demo** – MIDI Innovation Awards 2023 Finalist project: youtu.be/gkOqfOT8zXM.

Core Competencies

Music Information Retrieval & Audio ML: onset/beat tracking, chroma and key/chord detection, symbolic (MIDI) music modelling, DSP feature extraction (STFT, MFCC/LFCC/PLP/GFCC, S-transform, CQT), real-time and low-latency inference, edge/embedded deployment, benchmark and dataset design, human-centred evaluation of music systems.

Technical Stack

C++17, Python (TensorFlow, Keras, PyTorch, scikit-learn, Librosa), Elk Audio OS, JUCE, VST, OSC, MIDI, Raspberry Pi / ARM CPU profiling, AWS, SQL, ETL, Docker, CI/CD, Git.

Research and Work Experience

Independent Researcher | *Self-directed, Toronto, Canada* *Jan 2026 - Present*

- Authored two accepted papers: Smart Drums (JAES 2026, in press) and MIRaaS: Music Information Retrieval as a Service (IEEE IS2 2026, accepted).
- Maintain Nebula, a public C++17 audio-feature-extraction / MIR toolkit with per-feature ARM and Raspberry Pi latency benchmarks, 13 stars.
- AI-assisted development workflows using Claude Code for research tooling and documentation.

Postdoctoral Researcher | *University of Trento, Trento, Italy* *Jan 2025 - Dec 2025*

- Owned the full ML pipeline for streaming audio and IMU/gesture data – acquisition, DSP/feature engineering, training and evaluation, edge deployment, and monitoring – on the EU Horizon EIC Pathfinder project MUSMET.
- Published a real-time audio pattern detector achieving 14 ms inference on a Raspberry Pi 4 at F1 0.76 and 31.4% CPU, 74x faster than the 1038 ms DTW baseline it replaced (JAES 2026).
- Designed frozen-protocol benchmark suites and ablations quantifying accuracy/latency/CPU trade-offs.
- Built MIRaaS, a UDP edge-to-server offload architecture for music information retrieval, with a latency-characterization paper accepted at IEEE IS2 2026.

Visiting Researcher | *McGill University (IDMIL / CIRMMT), Montreal, Canada* *May 2024 - Aug 2024*

- Built a self-powered smart electric guitar, fusing IMU and audio signals via a hierarchical FSM to reach F1 0.78 on a 500-event corpus (IEEE IS2 2025).

Visiting Researcher | *University of Visual and Performing Arts, Colombo, Sri Lanka* *Jan 2024 - Mar 2024*

- Ran a human-centred evaluation of musicians' perception of smart musical instruments with embedded pattern detection (IEEE I3DA 2025).

Technical Consultant | *Forestpin (Pvt) Ltd, Colombo, Sri Lanka* *Aug 2020 - Feb 2021*

- Built ML and statistical pipelines for financial forensics, compliance monitoring, and risk detection on large structured enterprise data.

Research Engineer | *MAS Holdings (Pvt) Ltd, Colombo, Sri Lanka* *Jan 2015 - Jul 2020*

- Delivered end-to-end computer-vision and IoT systems for manufacturing at scale over 5.5 years, raising operational efficiency by up to 300%.
- Built real-time IoT ETL pipelines with C++/Python inference on embedded hardware; introduced CI/CD and code review practices for the engineering team.

Publications

Silva, N. & Turchet, L. (2026). Real-Time Audio Pattern Detection for Smart Musical Instruments. *Journal of the Audio Engineering Society*, 74(3). Silva, N. & Turchet, L. (2026, in press). Smart Drums. *Journal of the Audio Engineering Society*. Silva, N. & Turchet, L. (2026, accepted). MIRaaS: Music Information Retrieval as a Service. *IEEE IS2*. Silva, N. & Turchet, L. (2024). Real-Time Pattern Recognition of Symbolic Monophonic Music. *ACM Audio Mostly*. PhD thesis: "Embedded Real-time Musical Pattern Detection for Smart Musical Instruments," University of Trento, 2025.

Education

PhD, Information Engineering and Computer Science – University of Trento, Italy (2021 - 2025). Thesis: "Embedded Real-time Musical Pattern Detection for Smart Musical Instruments."
MSc, Telecommunication and Electronic Engineering – Sheffield Hallam University, UK (2016 - 2018).
BEng (Hons), Electronic Engineering – Sheffield Hallam University, UK (2013 - 2014).

Highlights

Author of four open-access music-content datasets on Zenodo (DoMP, DoPP, DoDP, DoDP2; 9,800+ recordings, 70+ musicians) and designer of MIRaaS, a music-information-retrieval-as-a-service architecture, giving direct, hands-on experience with the kind of audio-content pipelines a music and media company runs in production. MIDI Innovation Awards 2023 Finalist ("Hot Licks," Prototypes & Non-Commercial Software; covered by Sound On Sound and MusicRadar). Existing Montreal ties through a 2024 visiting research appointment at McGill University's IDMIL/CIRMMT lab. Fifteen-plus years as a performing and session guitarist, Trinity College London Rock & Pop Guitar Grade 8, bringing musician-informed intuition to music-content and audio-ML work.